Resistance-Aware Docking Prioritization of Antimalarial Leads from African Natural Products

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Abstract

Resistance-aware docking can help rank antimalarial candidates, but docking scores alone do not establish target engagement or mechanism of action. We assessed 17 candidates derived from African natural products across 136 PfDHFR and PfCRT wild-type and mutant systems using a docking-score retention metric (RRS). Among the 12 candidates with complete two-target data, the class distribution was A*: 1, A: 1, B: 4, C: 5, D: 1; network score and RRS were only weakly associated ($\rho = -0.2098$, adjusted $p = 1.0000$). Short 10 ns MD on four related but non-overlapping parent-study complexes left only PfCRT–214 with a bound ligand and an interpretable MM-GBSA estimate ($(-18.25 \pm 0.40)\,\mathrm{kcal\,mol^{-1}}$); a 16-system pilot on two Set-C candidates did not recover the docking retention pattern. An external panel of 39 ligands (312 records)

gave a mean RRS near 100, so mutant and wild-type scores were largely proportional under the same protocol. Within this computational panel the ranking is informative; it does not demonstrate biological polypharmacology or resistance circumvention.

Introduction

Drug resistance remains the principal threat to malaria control in Africa. An estimated 282 million cases and 610 000 deaths occurred globally in 2024, with the WHO African Region bearing the greatest burden[1]. Partial resistance to artemisinin derivatives, mediated by mutations in *Pfkelch13*, is now confirmed or suspected in at least eight African countries[2], and declining efficacy of some partner drugs has been reported[1,3]. Spatial-temporal modelling further indicates that the geographic footprint of *pfk13* markers in Africa is expanding[4], while genomic surveillance documents the co-occurrence of *Pfcrt* and *Pfdhfr* resistance variants across diverse transmission settings[5–7].

Polypharmacology is increasingly treated as a design strategy for complex disease biology, and multi-target-directed ligands represent an important translational class[8]. The rationale is evolutionary as well as pharmacological: under the single-drug–multi-target (SDMT) paradigm, simultaneous loss of activity at several relevant targets is proposed to raise the mutational barrier to treatment failure in a way that combination therapy alone does not[9]. That proposition is conditional, however. A list of predicted protein contacts is not a mechanism-of-action assignment; causal network analyses require an explicit perturbation model, a defined biological background, and evidence that the predicted interactions propagate to a phenotype[10]. Network-learning approaches to off-target prediction illustrate the value of graph context while underscoring that network inference remains distinct from experimental target engagement[11]. Natural products are attractive starting points because their stereochemical and scaffold complexity can support activity across multiple protein classes[12,13]. African natural products provide a particularly relevant chemical-space context, with resources such as ANPDB enabling explicit comparison with approved-drug and synthetic reference spaces[14].

In a previous study, we used generative molecular methods and dual-filter consensus docking (AutoDock Vina[15] and DiffDock-L[16]) to generate a library of 65 856 candidate molecules from 396 African natural products and 454 synthetic drugs[17]. We selected the present candidates against a panel spanning established resistance determinants and mechanistically distinct parasite processes: PfDHFR (dihydrofolate reductase, PDB 7F3Y) and PfCRT (chloroquine resistance transporter, PDB 6UKJ) represent clinically established resistance-associated contexts; PfATP4 (sodium efflux pump, PDB 9N10) probes ion homeostasis; and the 4GM2 entry represents PfClpR, not PfClpP, and is retained as a proteostasis-related parent-study system. This panel is deliberately a mechanistic cross-section rather than an exhaustive parasite targetome. It allows us to ask whether resistance-aware prioritization remains informative across distinct target classes, while avoiding the stronger claim that the selected compounds define a complete parasite-wide polypharmacology profile. PfCRT docking and MD were performed at physiological pH; results for this transporter are therefore conditional on that preparation and may differ under digestive-vacuole pH.

Rigid docking is efficient for ranking but treats the receptor and scoring landscape in a simplified manner. MD can test whether a docked pose remains compatible with protein flexibility, whereas endpoint methods such as MM-GBSA provide system-specific energetic summaries rather than calibrated thermodynamic observables[18,19]. This distinction is especially important for resistance analysis: a docking RRS is a hypothesis about relative mutant sensitivity, not a measurement of a mutant free-energy difference. Network pharmacology provides a complementary prioritization layer, but centrality is a topological weighting heuristic rather than a causal measure of target essentiality, consistent with the growing interest in multi-target antimalarial design more broadly[20]. We therefore treat these quantities separately: docking-derived RRS for mutant sensitivity, PNS for network-weighted target ranking, ACSI for chemical-space positioning, and targeted MD for post-docking structural scrutiny.

We asked whether target-level docking-score retention could distinguish predicted potency from predicted mutant sensitivity in a chemically diverse antimalarial library. The primary analysis used 12 candidates with eligible PfDHFR and PfCRT wild-type scores; five PfCRT-only candidates formed a coverage-limited sensitivity set. RRS was calculated from empirical Vina scores, whereas PNS and ACSI provided complementary ranking dimensions. Four parent-study wild-type complexes—chemically and numerically distinct from the 17 Set-C candidates—underwent 10 ns simulations as a methodological pose-retention check; a 16-system Set-C pilot provided the only trajectory-level comparison against docking-RRS for members of the prioritization cohort. Biological polypharmacology and resistance circumvention were not tested.

Methods

Study design

The primary analysis is docking-score retention (RRS) for the 12 Set-C candidates that have eligible PfDHFR and PfCRT wild-type scores. Three secondary analyses are reported separately: the five PfCRT-only candidates, a four-complex parent-study MD check, and a 16-system Set-C MD pilot. They address different coverage and sampling questions and are not combined with the primary RRS classes. The parent-study MD systems are not among the 17 Set-C candidates and do not validate the Set-C ranking.

Candidate selection and filtering

The Set-C cohort is a filtered prioritization cohort rather than a representative sample of the parent library. Class frequencies are therefore conditional on the stated MPO, SYBA, selectivity, multi-target, and scaffold-diversity filters and are not estimates for the full 65 856-molecule library.

Seventeen candidates were selected from the Set-C candidate panel generated from the

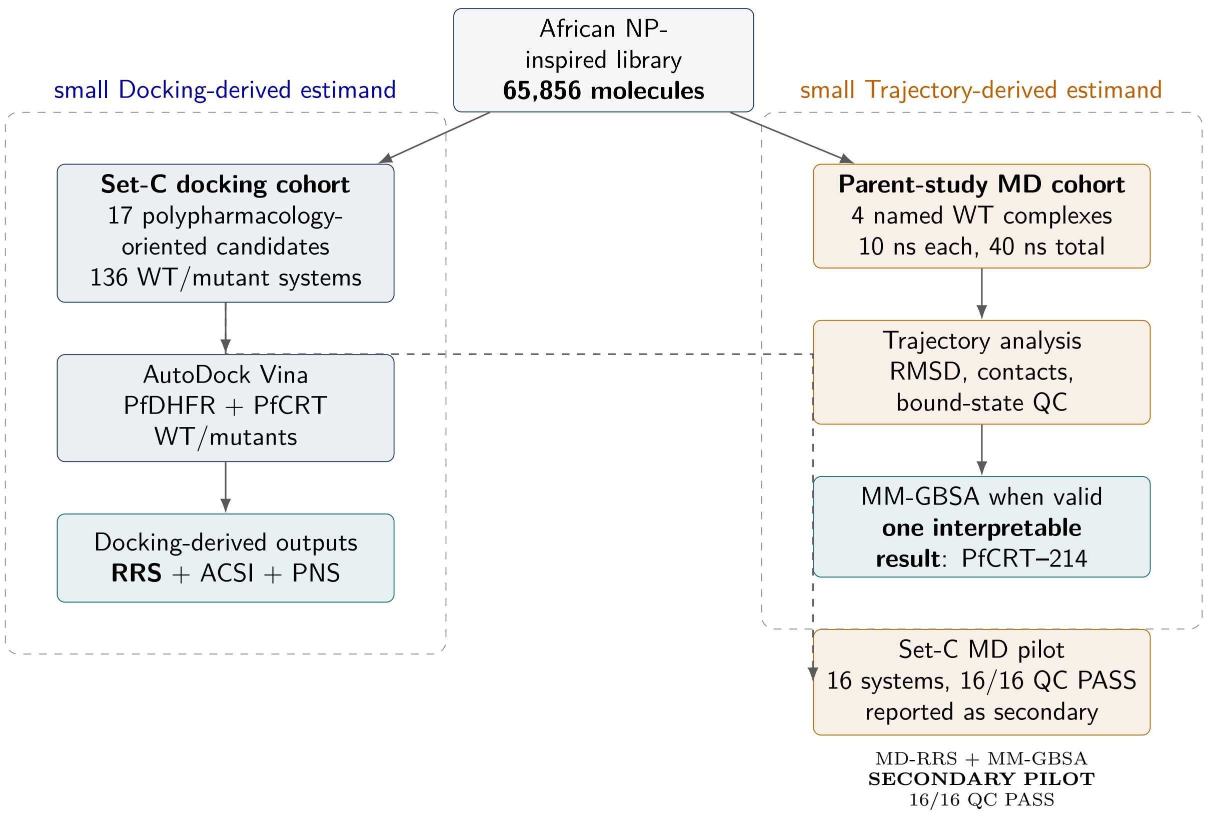

Figure 1: *Study design. Docking-derived RRS, ACSI, and PNS are computed on the 17 Set-C candidates. Four parent-study complexes, which do not overlap that list, are used only for a short MD and MM-GBSA check. The 16-system Set-C pilot (PP-01 and PP-02) is reported separately and does not define the RRS classes.*

library of 65 856 molecules described previously[17]. After MPO ≥ 0.70, SYBA > 0, and SI > 10 filtering, 19 913 primary leads remained; the present Set-C cohort is a further prioritization of that filtered set. The candidate-selection panel is distinct from the production-MD cohort. Three sequential filters were applied: (1) MPO score ≥ 0.70 (primary leads); (2) SYBA score > 0 (synthesisable); (3) selectivity index SI > 10 (selectively antiparasitic). The selection additionally required multi-target engagement of at least two of the four docking targets and scaffold diversity of no more than three compounds per Bemis–Murcko scaffold. The selection includes one of the previously named consensus leads (Ligand 87, PfDHFR). The four additional parent-study consensus leads (Ligands 201, 214, 438, 164) were not part of the 17-candidate docking set but were used to construct the wild-type complexes for the production MD check. The 164 system is identified as the 164-labelled PfClpR cohort; PDB 4GM2 is PfClpR rather than PfClpP, so this system is not structural validation of PfClpP. Five approved antimalarial drugs (artemisinin, pyrimethamine, chloroquine, lumefantrine, cipargamin) were included as reference controls for the docking analysis (Table 1).

Table 1: *Breakdown of the 136 docking protein–ligand systems by category (17 polypharmacological candidates $\times$ 8 wild-type/mutant PfDHFR and PfCRT structures). *Production MD was performed at* 10 ns *each (*40 ns *total) on 4 wild-type parent-study complexes distinct from the Set-C list. Ligand 164 is retained as the 164-labelled PfClpR cohort; PDB 4GM2 is PfClpR rather than PfClpP. The other parent-study systems were Ligand 201 (PfDHFR), Ligand 214 (PfCRT), and Ligand 438 (PfATP4).*

Category	Ligands	Targets	Systems	MD (ns)
PfDHFR WT + 4 mutants (Set-C)	17	5	85	—
PfCRT WT + 2 mutants (Set-C)	17	3	51	—
Docking total	–	–	**136**	—

The 136 docking systems were not subjected to production MD. The 16-system pilot and four non-overlapping parent-study complexes were analyzed separately.

Resistance mutation panel

We selected six resistance-associated states for comparison (Table 2); target identifiers are listed in the Supporting Information (Table S14). Each state corresponds to a WHO-listed molecular marker of antimalarial drug resistance, giving the panel a clinical anchor rather than an arbitrary residue choice[21].

Table 2: Resistance mutation panel.

Target	State	Resistance context	Evidence boundary
PfDHFR	N51I	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	C59R	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	S108N	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	I164L	High-level antifolate-resistance state	Comparator for severe resistance
PfCRT	K76T	Chloroquine-resistance marker	Transporter-resistance context
PfCRT	K76A	PfCRT resistance/comparator state	Not a prevalence estimate

Homology modelling of resistance mutants

Crystal structures of mutant targets were not available in the PDB. Homology models were generated with SWISS-MODEL using the wild-type structures as templates: PfDHFR (7F3Y) and PfCRT (6UKJ). PfATP4 and the 164-labelled parent-MD system were not part of the six-mutant homology-model panel; moreover, PDB 4GM2 is PfClpR rather than PfClpP. Model quality was assessed by QMEAN score, Global Model Quality Estimation (GMQE), Ramachandran analysis (MolProbity), and Cα RMSD to the wild-type template. Models were retained only when QMEAN exceeded -4.0, GMQE exceeded 0.7, Ramachandran favored residues exceeded 95 %, and backbone RMSD was below 1.5 Å. All six mutant models (PfDHFR N51I, C59R, S108N, I164L; PfCRT K76T, K76A) met these thresholds and were retained as docking receptors; a summary is given in the Supporting Information (Table S2). K76A was included as a charge-neutralising isosteric perturbation of the canonical K76T substitution, allowing the contribution of the charged Lys76 anchor to be examined separately from the threonine substitution. Model scores are not experimental structures, and the lack of mutant crystal structures or replicate ensembles remains a source of uncertainty.

Docking protocol assessment

The docking protocol and V2-corrected grid parameters used here are identical to those established in the parent study[17]: AutoDock Vina 1.2.7[15] with exhaustiveness 64 against grids centered on the corrected V2 binding-site coordinates of PfDHFR (7F3Y), PfCRT (6UKJ[22]), PfATP4 (9N10), and the historically labelled PfClpR/4GM2 entry (not PfClpP). As an independent boundary audit of these fixed boxes, P2Rank[23] pocket predictions were compared with the grid centers; the concordant PfCRT result and the frame caveats for the other receptors are presented in the Supporting Information (Figure S1).

Parent-study enrichment benchmarks (V2 grids) are summarized here and tabulated in full in the Supporting Information (Table S1). On the MMV Malaria Box consensus (Vina+DiffDock), ROC-AUC was 0.924 (PfDHFR), 0.971 (PfCRT), and 1.000 (PfATP4). In contrast, DEKOIS 2.0 enrichment for PfDHFR with Vina alone was null (ROC-AUC 0.450, $EF_{5\%} = 0.00$), indicating that score-based recovery of known actives is protocol- and library-dependent. Redocking was considered accurate when heavy-atom RMSD was < 2.0 Å relative to the crystal pose; the MTX redock attempt was a clear failure (RMSD ≈ 30 Å). AutoDock Vina was used for all reported Set-C scores; GNINA yielded no usable score for this panel.

Secondary docking calibration

A secondary Tartarus cross-docking analysis used QuickVina on the primary-lead panel. The composite MPO–affinity association was null and exploratory; complete correlations are provided in the Supporting Information (Tables S8 and S9).

MPO sensitivity and predicted ADMET profile

MPO-weight sensitivity and single-source ADMET-AI predictions were used descriptively; the complete sensitivity and endpoint results are provided in the Supporting Information

(Figure S2 and Table S3). These predictions were not experimentally validated.

Resistance resilience score definition

All 17 candidates were docked against each wild-type and mutant structure using AutoDock Vina 1.2.7 with exhaustiveness 64. The 136 systems were ranked with empirical Vina scores rather than calibrated binding free energies. Grid boxes were centered on the co-crystallized ligand position with 25 Å dimensions, matching the audited V2 configuration files.

For target t, let $s_{\mathrm{Vina},i,\mathrm{WT},t}$ and $s_{\mathrm{Vina},i,m,t}$ denote the signed Vina scores. Candidates with $|s_{\mathrm{Vina},i,\mathrm{WT},t}| < 5.0\,\mathrm{kcal\,mol^{-1}}$ are excluded from that target's score-retention analysis. The target-specific Resistance Resilience Score (RRS) is

$$\mathrm{RRS}_{i,m,t} = \frac{|s_{\mathrm{Vina},i,m,t}|}{|s_{\mathrm{Vina},i,\mathrm{WT},t}|} \times 100. \tag{1}$$

RRS is a relative docking-score-retention statistic, not a free-energy difference or biochemical resistance measurement. The available-target summary averages the defined target-specific values across eligible targets and mutant states. To avoid letting unequal target coverage drive the main ranking, the primary analysis is restricted to the 12 candidates with eligible PfDHFR and PfCRT wild-type scores; the five PfCRT-only candidates are reported separately.

The mutually exclusive classes are defined from the available mutant values; primary class counts use the complete two-target subset:

- **Class A***: all available mutant RRS values are at least 80 % and the minimum eligible wild-type score magnitude is at least $7.0\,\mathrm{kcal\,mol^{-1}}$.
- **Class A**: all available mutant RRS values are at least 80 %, but the minimum eligible wild-type score magnitude is below $7.0\,\mathrm{kcal\,mol^{-1}}$.
- **Class B**: all available mutant RRS values are at least 70 %, but at least one is below

80 %.

- **Class C**: at least one available mutant RRS value is at least 80 %, but not all available values reach 70 %.
- **Class D**: no available mutant RRS value reaches 80 %.

The classes are mutually exclusive. The cut-offs follow the distribution of Vina scores in this panel and have not been calibrated to biochemical resistance or clinical outcome. Sensitivity to alternative thresholds is reported in the Supporting Information (Table S6).

Secondary ranking descriptors

ACSI places each candidate relative to approved-drug and African natural-product reference spaces; weight-sensitivity checks are in the Supporting Information (Table S4). PNS is a network-weighted ranking score; PfCRT centrality was imputed, with sensitivity reported in the Supporting Information (Table S7). Associations among metrics were assessed by Spearman rank correlation with 100 000 permutations, Bonferroni adjustment for three tests, and bootstrap intervals from 10 000 resamples. At $n = 12$ these correlations are descriptive and are not powered tests of independence.

MD protocol

We simulated all four parent-study wild-type systems for 10 ns with CHARMM36m[24]/GAFF2[25] at 310.15 K using GROMACS[26,27]. The 10-ns length is intended as a pose-retention and local-geometry check, not as a measurement of residence time or converged free energy. We analyzed the trajectories after removing periodic-boundary discontinuities. MDAnalysis[28] was used to calculate contacts, H-bonds, RMSF, and radius of gyration every 10 frames and RMSD on every frame.

MM-GBSA endpoint estimates were obtained from production trajectories using the GB$^{\mathrm{OBC2}}$ model (igb=5) with 0.150 M salt concentration, via the `gmx_MMPBSA` v1.5.0.3[18]

CHARMM36-to-AMBER conversion. All MM-GBSA values are endpoint enthalpic scores from a single-trajectory protocol; configurational entropy is not included. Uncertainty is reported as the standard error of the mean over stored snapshots (101 for the interpretable PfCRT–214 system; 100 for Set-C pilot systems). Within-trajectory and propagated standard deviations are tabulated in the Supporting Information and are not replicate uncertainties.

The 16-system Set-C pilot used the same temperature and length, with ligand parameters from OpenFF 2.2.0 (AM1-BCC) rather than GAFF2. Absolute MM-GBSA values are therefore not compared between the two cohorts. Each mutant was run as a single replicate; trajectories were accepted for endpoint analysis only after geometric quality checks.

Results

Primary docking-RRS classification

Mean target-specific RRS was 75.1 for PfDHFR (12 candidates) and 86.2 for PfCRT (17 candidates); full target-wise summaries are in the Supporting Information (Table S5). The difference is in docking-score retention between panels, not evidence of different biological resistance.

On the complete two-target set ($n = 12$) the classes were A*: 1, A: 1, B: 4, C: 5, and D: 1. When all 17 candidates are classified on whatever targets are eligible, the counts become A*: 5, A: 1, B: 5, C: 5, and D: 1; the five PfCRT-only compounds cannot be compared on equal footing with the two-target set. Per-compound RRS profiles across all six mutants are shown in Figure S3. Varying the wild-type eligibility threshold from $4.0\,\mathrm{kcal\,mol^{-1}}$ to $5.0\,\mathrm{kcal\,mol^{-1}}$ left class counts unchanged; raising retention thresholds from 70 %/80 % to 80 %/90 % increased Class D from 1 to 9 (Table S6).

Table 3: *Target-specific docking-score retention for the* 17 *Set-C candidates. RRS is computed separately for each target and only when the corresponding wild-type Vina score magnitude is at least* 5.0 kcal mol^{-1}*. A dash denotes an ineligible target, not a zero score. The primary class applies only to the* 12 *complete two-target candidates; the available-target class is shown for all* 17*. Classes are mutually exclusive: A* requires all available mutant RRS values* $\geq 80\,\%$ *and the minimum eligible wild-type score magnitude* ≥ 7.0 kcal mol^{-1}*; A has the same RRS criterion but a weaker minimum wild-type score; B requires all values* $\geq 70\,\%$ *but not all* $\geq 80\,\%$*; C requires at least one value* $\geq 80\,\%$ *but not all values* $\geq 70\,\%$*; D is the residual class in which no available mutant reaches* 80 %*. All scores are empirical AutoDock Vina scores; RRS is not a free-energy or resistance measurement.*

Candidate	WT PfDHFR	WT PfCRT	Eligible targets	N51I	C59R	S108N	I164L	K76T	K76A	Primary class	Available class
PP-01	7.5	9.3	both	86.4	85.5	87.6	83.9	92.5	90.9	A*	A*
PP-02	–	9.1	PfCRT	–	–	–	–	87.9	94.6	–	A*
PP-03	10.3	11.3	both	68.1	68.3	65.8	69.0	80.0	81.4	C	C
PP-04	8.0	7.0	both	68.8	66.9	68.8	73.5	105.3	103.6	C	C
PP-05	–	10.5	PfCRT	–	–	–	–	77.1	79.3	–	B
PP-06	–	9.9	PfCRT	–	–	–	–	90.6	94.1	–	A*
PP-07	7.6	9.1	both	71.8	71.8	72.4	72.2	84.2	77.6	B	B
PP-08	7.6	7.4	both	62.1	62.1	67.2	66.3	83.9	83.1	C	C
PP-09	7.1	8.6	both	70.6	73.0	75.8	76.5	75.8	73.5	B	B
PP-10	7.7	8.4	both	76.0	76.4	75.7	75.8	89.4	89.6	B	B
PP-11	–	10.0	PfCRT	–	–	–	–	82.7	82.5	–	A*
PP-12	7.5	7.8	both	75.2	66.1	62.9	72.5	70.4	87.2	C	C
PP-13	–	8.3	PfCRT	–	–	–	–	110.2	99.4	–	A*
PP-14	7.6	7.5	both	62.2	65.5	68.0	64.9	78.8	77.7	D	D
PP-15	5.6	9.3	both	123.2	112.7	125.9	131.8	86.2	90.1	A	A
PP-16	7.6	8.7	both	72.9	74.7	73.7	75.9	80.1	80.6	B	B
PP-17	8.3	7.4	both	59.3	61.2	59.5	58.9	76.8	93.4	C	C

Chemical-space and network descriptors

Two of 17 candidates had ACSI above 0.70 (cohort mean 0.543). ACSI measures proximity to reference chemical space, not activity or resistance. PNS rankings and imputation details are in the Supporting Information (Table S7).

Cross-metric associations

Table 4: *Cross-metric Spearman correlations on the complete two-target set (*$n = 12$*). Permutation p-values use* 100 000 *permutations; adjusted p-values use Bonferroni correction for three tests. Bootstrap intervals are percentile* 95 % *intervals from* 10 000 *resamples. At this sample size the correlations are descriptive.*

Metric pair	Hypothesis	Spearman ρ	p_{perm}	p_{adj}	Bootstrap 95 % interval
PNS vs. RRS	H1	−0.209 8	0.514 4	1.000 0	[−0.7582, 0.5429]
ACSI vs. RRS	H2	−0.405 6	0.192 2	0.576 6	[−0.8917, 0.2857]
RRS vs. weakest WT score	H3	−0.166 1	0.603 8	1.000 0	[−0.7300, 0.5932]

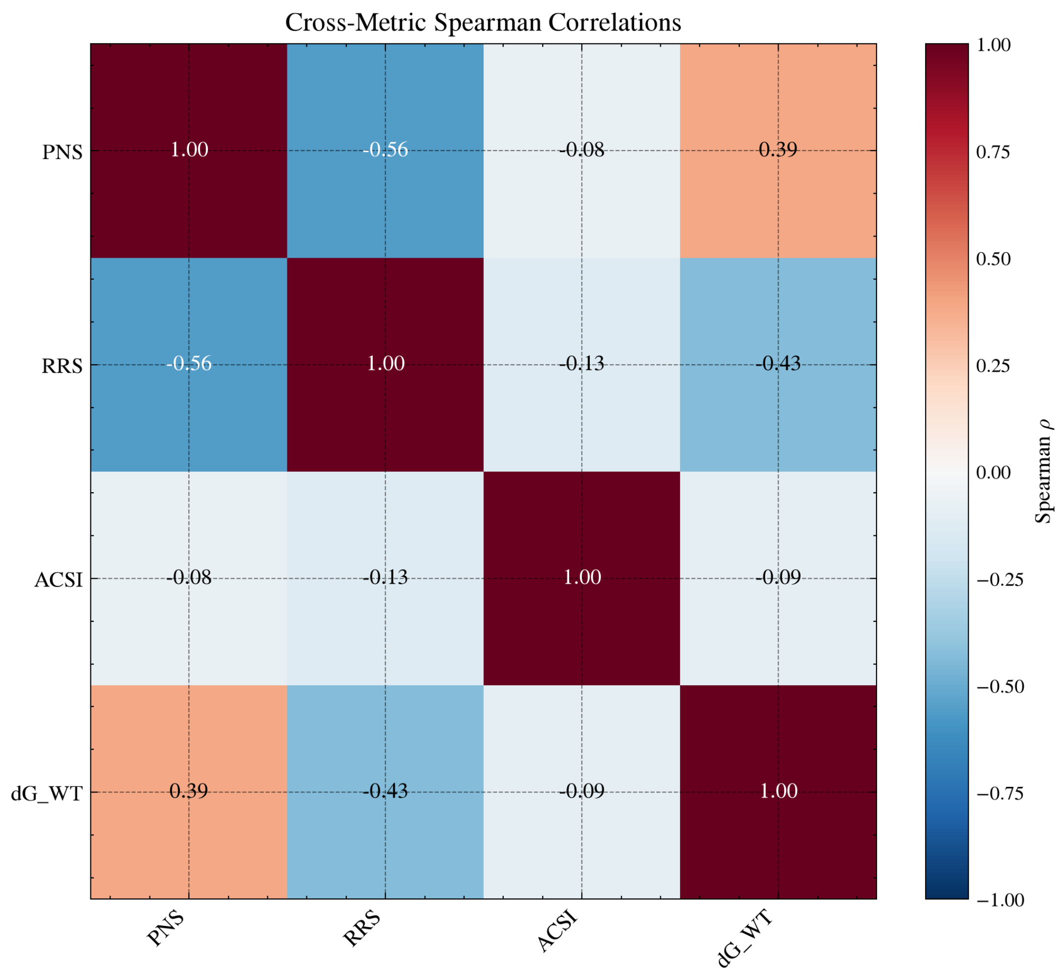

Figure 2: *Cross-metric Spearman correlations for Set-C candidates. The primary analysis uses the 12 complete two-target candidates; the 17-candidate matrix is shown only for comparison when target coverage is incomplete. PNS already includes wild-type docking scores, so that pair is not an independent test.*

Table 4 reports the complete two-target set. On all 17 candidates, PNS–RRS was $\rho = -0.5588$ (permutation $p = 0.0222$, adjusted $p = 0.0667$), ACSI–RRS $\rho = -0.1324$ ($p = 0.6117$), and RRS versus weakest wild-type score $\rho = 0.1759$ ($p = 0.4947$), but unequal target coverage limits interpretation of the full set.

On the two-target set, PNS and RRS were only weakly associated ($\rho = -0.2098$, adjusted $p = 1.0000$), as were ACSI and RRS ($\rho = -0.4056$, adjusted $p = 0.5766$). RRS was not associated with the weakest eligible wild-type score ($\rho = -0.1661$, adjusted $p = 1.0000$). PNS already includes wild-type docking scores, so its correlation with those scores is not an independent test. The bootstrap intervals are wide, as expected for $n = 12$.

Computational prioritization intersection

Requiring Class A or A*, ACSI > 0.5, and above-mean PNS leaves a single candidate (PP-15) on the two-target panel. That intersection is a shortlist for experiment, not evidence of activity or resistance resilience.

Parent-study MD check (non-overlapping cohort)

These four complexes are not among the 17 Set-C candidates. The trajectories test whether the docked poses remain bound on a short timescale; they do not validate the Set-C ranking.

Table 5: *Production MD stability metrics for parent-study wild-type complexes (10 ns). BB RMSD, ligand RMSD, minimum distance, contacts, and H-bonds are computed from PBC-unwrapped trajectories.*

Target	Atoms	BB RMSD (Å)	Lig. RMSD (Å)	Min dist (Å)	Contacts	H-bonds	Interpretation
PfClpR-labelled cohort (164)	43 035	1.31	1.11	67.42	0.0	23.7	Unbound
PfDHFR (201)	134 153	3.05	1.70	78.21	0.0	21.5	Unbound
PfCRT (214)	76 920	4.47	2.45	3.19	78.1	23.1	Bound
PfATP4 (438)	420 801	12.27	2.13	2.25	178.4	47.7	Bound

Two systems, PfCRT and PfATP4, retained their ligands throughout production. PfCRT had a backbone RMSD of 4.47 Å, a ligand RMSD of 2.45 Å, 78.1 contacts, and 23.1 H-bonds (minimum heavy-atom distance 3.19 Å). PfATP4 shows a backbone RMSD of 12.27 Å, a ligand RMSD of 2.13 Å, 178.4 contacts, and 47.7 H-bonds (minimum distance 2.25 Å). The high backbone RMSD indicates substantial structural drift in this model. Because the simulation omitted a membrane and the topology conversion was not valid for MM-GBSA, the apparent ligand persistence should not be interpreted as validated binding thermodynamics. PfCRT ligand 214 anchors to Lys34 and engages in a robust H-bond network (Gln101, Thr30, Gln297) and a hydrophobic core (Leu301, Leu105, Ala33), consistent with a stable pose in this trajectory; neither specificity nor target engagement is established.

The other two systems, the 164-labelled PfClpR cohort and PfDHFR, had stable protein conformations (backbone RMSD 1.31 Å and 3.05 Å, respectively) but the ligands are completely unbound (minimum distances 67.42 Å and 78.21 Å, zero contacts). This indicates either incorrect initial placement or rapid dissociation during equilibration, not a PBC artefact. Consequently, MM-GBSA analyses for these systems are not interpretable as affinity estimates. Protein equilibration alone was insufficient to retain the docked poses.

Table 6: *Endpoint interpretation for four parent-study complexes used for production MD. These systems are distinct from the 17 Set-C candidates. Only PfCRT–214 yields an interpretable MM-GBSA endpoint; dissociated systems are reported as N/A.*

Ligand	Target	ΔG_{bind} (kcal mol^{-1})	SEM	Interpretation
214	PfCRT-WT	−18.25	0.40	Interpretable; ligand bound†
164	PfClpR-labelled cohort-WT	–	–	N/A; ligand dissociated‡
201	PfDHFR-WT	–	–	N/A; ligand dissociated§
438	PfATP4-WT	–	–	N/A; conversion corrupted*

†Ligand remains bound (minimum distance 3.19 Å, 78 contacts); conversion via `gmx_MMPBSA` native converter.
The quoted uncertainty is the standard error of the mean over the 101 stored snapshots (see Methods).
‡Ligand dissociates to 67.4 Å minimum distance with zero contacts during production; excluded from binding-mode conclusions.
§Ligand dissociates to 78.2 Å minimum distance with zero contacts; same caveat applies.
*Excluded: conversion artifact in the two-chain PfATP4 topology (+473 kcal mol^{-1} van der Waals inflation); structural persistence is reported separately.

Only the PfCRT–ligand 214 system yields an interpretable MM-GBSA endpoint estimate ($\Delta G_{\text{bind}} = (-18.25 \pm 0.40)$ kcal mol^{-1}). The 164-labelled and PfDHFR–201 ligands dissociated during production; the PfATP4 system was excluded because CHARMM36-to-

Figure 3: *Detailed stability analysis for the parent-study PfCRT–Ligand 214 complex over* 10 ns *production MD. The system exhibits transporter flexibility while retaining a bound ligand; cohort-level stability values are reported in Table 5.*

AMBER conversion of its two-chain topology produced a systematic van der Waals artifact ($+473\,\mathrm{kcal\,mol^{-1}}$). No Vina–MM-GBSA correlation is estimated across the parent-study and Set-C cohorts.

Set-C MD pilot (secondary)

A 16-system pilot on PP-01 and PP-02 asked whether short trajectories and MM-GBSA would recover the docking-RRS pattern. All 16 trajectories met the geometric acceptance checks. For the eight mutants with a docking wild-type reference, docking predicted weaker mutant scores, while trajectory metrics more often indicated retained or tighter contacts (seven of eight comparisons disagreed in direction). MM-GBSA mutant/wild-type ratios exceeded 100 % for eight of twelve mutants; those ratios are endpoint comparisons, not free-energy measures of resilience. The pilot therefore does not reproduce docking-RRS and does not imply resistance or tighter binding. Full values are in the Supporting Information (Figure S5 and Table S16).

As a bounded consistency check, we intersected the docking-RRS retention criterion (class-A, RRS $\geq 80\,\%$) with pilot MD-RRS retention parity (mean minimum-distance ratio < 100) at pilot scope. Seven of twelve gate-eligible rows passed both gates: PP-01 satisfied the two-target gate (PfCRT via K76A, PfDHFR in four of four mutants), whereas PP-02 did not, because its PfDHFR states lack a docking wild-type anchor under the $5.0\,\mathrm{kcal\,mol^{-1}}$ eligibility rule and only one target was gate-eligible. The single row above parity (PP-01 PfCRT K76T, 102.7) lies within the $2.0\,\mathrm{kcal\,mol^{-1}}$ endpoint noise floor and is read as retention, not gain (Table S19).

A single-ligand feasibility probe (PP-15, outside the Set-C cohort) extended the same OpenFF protocol to wild-type PfDHFR and PfCRT: Vina best poses were $-8.62\,\mathrm{kcal\,mol^{-1}}$ and $-7.81\,\mathrm{kcal\,mol^{-1}}$, respectively, and both systems completed 10 ns production trajectories with valid checkpoints, yielding MM-GBSA estimates of $(-29.52 \pm 0.33)\,\mathrm{kcal\,mol^{-1}}$ (PfDHFR) and $(-27.79 \pm 0.49)\,\mathrm{kcal\,mol^{-1}}$ (PfCRT); as for the pilot, no RRS estimate is reported for this probe (Table S18).

Table 7: *Set-C pilot MM-GBSA estimates (exploratory; not used to assign RRS classes).* ΔG_{bind} *is the mean over* 100 *snapshots* $\pm$ *SEM; within-trajectory standard deviations are in the Supporting Information. MMG-RRS is the mutant/wild-type endpoint ratio (*$\times$100*; wild type =* 100*). Docking RRS is shown for comparison (n/a: no wild-type docking reference for PP-02 PfDHFR). Single-replicate endpoints should be read only as within-protocol contrasts.*

Candidate	Target	State	ΔG_{bind} (kcal mol^{-1})	SEM	MMG-RRS	Docking RRS
PP-01	PfCRT	K76A	−35.29	1.17	115.3	90.86
PP-01	PfCRT	K76T	−29.51	2.65	96.4	92.47
PP-01	PfCRT	WT	−30.61	1.65	100.0	–
PP-01	PfDHFR	C59R	−26.56	1.91	105.0	85.47
PP-01	PfDHFR	I164L	−26.13	2.34	103.3	83.87
PP-01	PfDHFR	N51I	−29.83	1.50	118.0	86.40
PP-01	PfDHFR	S108N	−30.21	3.62	119.5	87.60
PP-01	PfDHFR	WT	−25.29	2.94	100.0	–
PP-02	PfCRT	K76A	−24.53	1.27	97.8	94.62
PP-02	PfCRT	K76T	−30.45	1.38	121.4	87.91
PP-02	PfCRT	WT	−25.08	2.56	100.0	–
PP-02	PfDHFR	C59R	−28.85	1.88	95.7	n/a
PP-02	PfDHFR	I164L	−31.06	1.52	103.0	n/a
PP-02	PfDHFR	N51I	−34.02	2.68	112.8	n/a
PP-02	PfDHFR	S108N	−27.09	1.55	89.8	n/a
PP-02	PfDHFR	WT	−30.16	2.06	100.0	–

External docking replication

Docking of 39 external ligands across the same eight PfDHFR/PfCRT states gave 312 finite scores. Bootstrap mean RRS was 100.45 (99.59–101.32) and the Class-A fraction was 0.974 (0.923–1.000). A mean near 100 means mutant and wild-type Vina scores were on average almost proportional, so this panel adds little mutant/wild-type contrast under the same protocol. It is a computational sensitivity check, not validation of the Set-C ranking or of experimental resistance (Table S19).

Discussion

Resistance-resilient design

RRS compares each mutant score with the corresponding wild-type score after excluding wild-type magnitudes below 5.0 kcal mol^{-1}. It is a docking-derived prioritization statistic, not a biochemical measurement of resistance. The classes combine docking-score magnitude with relative score retention (Table 3). Class A* combines a wild-type score magnitude of at least $7.0\,\mathrm{kcal\,mol^{-1}}$ with RRS values of at least 80 % across all tested mutants. It is a prioritization tier, not evidence of resistance circumvention. Classes B and C identify progressively narrower resilience profiles, while Class D denotes that no available mutant reaches the 80 % retention threshold. These categories can be aligned with regional resistance surveillance in a future decision analysis, but they do not justify clinical deployment or therapeutic recommendations on their own[29].

This distinction is relevant in the current epidemiological context. The 2025 World Malaria Report documents declining efficacy of some artemisinin partner drugs alongside the spread of *Pfkelch13*-mediated partial resistance, with both trends concentrated in the African Region[1]. Resistance-aware prioritization can therefore be a useful design criterion rather than a guarantee of clinical relevance: compounds selected without an explicit mutant-sensitivity analysis may be poorly matched to the resistance states that are relevant in a given parasite population, but computational scores alone cannot establish that mismatch or its therapeutic consequence.

The external docking panel (mean RRS ≈ 100) underscores the same boundary. Under identical scoring and mutation definitions, mutant and wild-type Vina scores remained nearly proportional on an independent ligand set. Possible explanations include limited sensitivity of the Vina score to these particular mutations in that chemical space, insufficient structural perturbation captured by homology models, or genuine score retention for many of those ligands. The data do not distinguish among these alternatives; they caution against treating RRS as a calibrated resistance predictor outside the primary cohort.

An independent scoring function speaks to one of these alternatives. Rescoring the same 312 poses with the GNINA CNN affinity (post-processing only) reproduced the class-level picture: all 38 eligible ligands were class A under both scoring layers, with no class discordance (Table S19). The retention-not-gain pattern is therefore not an artefact of a single scoring function. The ligand-level rank correlation between layers was moderate ($\rho = 0.558$) and per-mutant correlations were weak or null (e.g. C59R -0.076), so the consensus supports class-level retention but not per-mutant rank transfer across scoring functions. Bootstrap intervals (1×10^4 resamples) bound the same conclusion on the primary cohort: the class- A* fraction was 0.294 (0.118–0.529) and the PfCRT per-mutant retention intervals excluded 100 (K76T 85.4, 81.0–90.4; K76A 87.0, 83.1–90.9), whereas the PfDHFR intervals were wide and overlapped 100 for I164L (Table S19).

The relationship between RRS and molecular topology remains unresolved. The companion TDA analysis reported associations in separate overlapping cohorts[30]; because those candidates share provenance with the present study, the estimates are not independent validation. Topological descriptors should therefore be tested prospectively in larger, matched panels with mutant-state free-energy or experimental measurements before being used as a design rule.

The PfCRT component of this picture is mechanistically plausible in light of long-timescale simulations of the chloroquine resistance transporter: 130 µs of atomistic MD across CQ-sensitive and CQ-resistant isoforms established that Lys76 physically obstructs the drug-binding site while serving as the electrostatic anchor for peptide substrates, with R371I and Q271E acting in concert with K76T to open drug access[31]. Our mutant-vs-wild-type end-point contrasts are consistent with this charged-cavity picture, in which perturbing residue 76 reshapes the local electrostatics that gate both ligand retention and substrate specificity. This consistency is contextual support, not an independent validation of the present docking scores.

Chemical and network context

ACSI places candidates relative to approved-drug and African natural-product chemical space. The mean was 0.543, and two of 17 candidates (11.8 %) exceeded 0.70. PNS is a network-weighted rank; on the two-target set it associated only weakly with RRS ($\rho =$

-0.2098, adjusted $p = 1.0000$). At $n = 12$ the correlations mainly describe how the ranks line up in this cohort; they are not a strong test that the metrics are independent. ACSI, PNS, and RRS remain complementary ranking tools, not independent measures of biology, and their meaning depends on the chosen reference spaces, network, and targets[11,13,14].

Role of structural follow-up

Short MD on the parent-study complexes shows both the value and the limit of structural follow-up when the compounds are not the ranked Set-C set. Two of four systems kept a bound ligand; the 164-labelled PfClpR and PfDHFR–201 systems did not. Only PfCRT–214 passed the checks needed for MM-GBSA, giving $(-18.25 \pm 0.40)\,\mathrm{kcal\,mol^{-1}}$—a system-specific endpoint that cannot calibrate Set-C scores. As is standard for endpoint methods, that number is a within-protocol guide, not a converged free energy[18,19].

The Set-C pilot is the only trajectory comparison with docking-RRS for compounds that are actually on the prioritization list. Mutant MM-GBSA did not recover the docking-predicted weakening, so whether RRS tracks mutant free-energy resilience is still open. With one replicate and 10 ns per system, the pilot speaks to local geometry and endpoint consistency, not residence time or experimental resistance. The pilot-scope retention gate illustrates the same boundary at class level: PP-01, the only candidate with a docking wild-type anchor on both targets, satisfied the two-target gate (docking RRS $\geq 80\,\%$ together with pilot MD-RRS below parity), whereas the directional mismatch observed in most mutant–wild-type pairs persists. Because the pilot bound fraction is saturated at $100\,\%$ for all sixteen systems, the gate is a coarse consistency check, not an affinity-equality proof.

Toward experimental validation. None of the scores reported here is biochemical or biophysical evidence. Useful next steps would be matched IC_{50} measurements of the highest-RRS candidates on wild-type and mutant PfDHFR; PfCRT binding or transport assays at digestive-vacuole pH; and short growth-inhibition tests in culture to see whether multi-target computational signals translate to a measurable phenotype. Agreement or disagreement with experiment would show which computational signal, if any, is useful for each target. The multi-target, mutation-resilient candidates prioritised here are also natural inputs to machine-learning predictors of antimalarial combination synergy, in which the same molecular features would be scored for pairwise combination effects[32]; such synergy predictions remain outside the scope of the present study.

Comparison with existing polypharmacology and resistance-aware approaches

Conventional multi-target docking ranks compounds by aggregated wild-type scores and does not treat mutant-state retention as a primary quantity. Network pharmacology often weights targets by centrality without comparing wild-type and resistance-associated structures under one scoring protocol[11]. Under the SDMT frame, the present class-A* readout combines exactly the two axes such a strategy requires—target breadth in the four-target wild-type matrix and per-target mutational retention in the RRS panel—and is therefore reported as an SDMT-style hypothesis, not as demonstrated multi-target engagement[9]. Free-energy methods such as FEP or TI can report mutant binding differences with clearer physical content, but they remain costly for early triage and need reliable structures and force fields. The approach used here sits between those extremes: RRS is an explicit mutant/wild-type docking-score ratio, while ACSI, PNS, and short MD remain separate layers rather than substitutes for one another. None of them replaces IC_{50} panels, transporter assays, or converged free-energy calculations; they are intended only to shortlist compounds for those experiments[20,30].

Limitations

Mutant receptors were homology models. All six met the QMEAN, GMQE, Ramachandran, and RMSD thresholds, yet model quality is not experimental accuracy. MD was limited to

10 ns on four parent-study complexes and a single-replicate 16-system Set-C pilot; neither design addresses long residence times or convergence. An independent PP-01 PfCRT wild-type run differed from the reference endpoint by 2.01 $\mathrm{kcal\,mol^{-1}}$, which we take as a rough noise floor given snapshot autocorrelation (Table S17). Single-replicate mutant endpoints, including the Set-C MM-GBSA values, are sensitive to frame choice and periodic-boundary handling. MM-GBSA omits configurational entropy; only PfCRT–214 satisfied the bound-state and topology requirements. The four-target panel is not a full parasite targetome.

RRS class frequencies depend on the selection filters and on unequal target coverage; they should not be extrapolated to the full library. PfCRT network centrality was imputed, so PNS depends on how the network is built (Table S7). Network-pharmacology scores are generally sensitive to database coverage and hub bias[33]; the PNS reported here is therefore a bounded ranking heuristic, not a validated network property. Force fields differed between the parent-study and Set-C cohorts, so absolute MM-GBSA values are not compared across protocols. Cross-metric correlations rest on $n = 12$ complete two-target candidates and have wide intervals. PfCRT was prepared at physiological pH rather than digestive-vacuole pH, which can change rankings[17]. Five candidates lacked an eligible PfDHFR wild-type score and appear only in the PfCRT-only analysis, a consequence of receptor setup rather than demonstrated selectivity.

Conclusion

We ranked antimalarial candidates from African natural-product space with docking-based RRS kept separate from chemical-space, network, and short MD scores. On the 12 candidates with complete PfDHFR and PfCRT data the RRS classes were A*: 1, A: 1, B: 4, C: 5, and D: 1. Correlations among metrics were weak at $n = 12$. On related but non-overlapping complexes, short MD retained a bound ligand in two of four systems and gave an interpretable MM-GBSA value only for PfCRT–214; a Set-C pilot did not match docking-RRS direction.

An external docking panel averaged near RRS 100, with little mutant/wild-type contrast under the same protocol. The ranking is useful inside this computational panel. It does not show biological polypharmacology, a systems-level mechanism, or experimental resistance circumvention. Target engagement assays, phenotypic tests, longer or free-energy simulations on mutants, and prospective experiments remain necessary.

Authors' Contributions

Myke Vital Sao Temgoua: Methodology, software, formal analysis, investigation, data curation, writing – original draft. Jean-Pierre Tchapet Njafa: Methodology, software, supervision, writing – review & editing. Penabei Samafou: data curation, writing – review & editing. Wilfred Fon Mbacham: conceptualization, validation, writing – review & editing. Serge Guy Nana Engo: conceptualization, methodology, supervision, writing – review & editing. All authors read and approved the final manuscript.

Data Availability

Docking results, metric outputs, analysis scripts, and the associated source-data records are available at `https://github.com/NanaEngo/Malaria_codesV2` under the MIT licence. A versioned snapshot of this repository will additionally be archived on Zenodo to provide a permanent, citable DOI concurrent with publication. The large parent-study trajectories are not included in this article's data record; the conclusions drawn here are restricted to the trajectory-derived summaries and binding estimates explicitly reported above.

Competing Interests

The authors declare no competing interests.

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Use of Artificial Intelligence

AI assistants were used to help with code and analysis scripts. All AI-assisted text and code were reviewed and edited by the authors, who take full responsibility for the article. No generative AI tool is an author. Computational results, statistics, and quantitative claims were produced and checked by the authors.

Supporting Information Available

Docking enrichment and homology-model quality tables; P2Rank box audit; MPO, ADMET, and ACSI sensitivity; RRS radar profiles and threshold sensitivity; PNS imputation and Tartarus calibration; cohort and evidence-scope tables; PlasmoDB identifiers; parent-study and Set-C pilot MD/MM-GBSA summaries; robustness and Paper 1 context tables (PDF).

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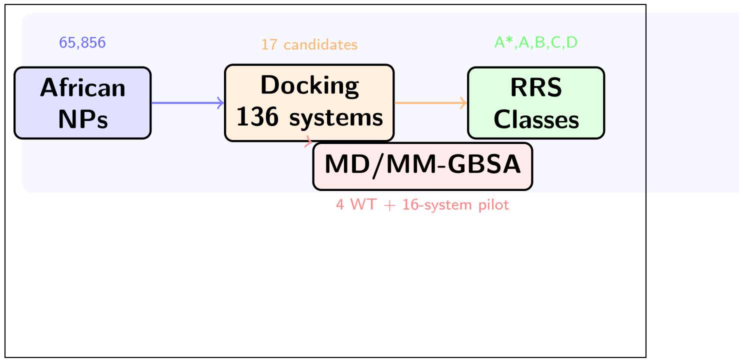
TOC Graphic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A*,A,B,C,D
RRS Classes
MD/MM-GBSA
4 WT + 16-system pilot